

Machine learning-based emissions monitoring for EU ETS compliance in energy-intensive manufacturing, leveraging ultra-precise detection to meet tightened benchmark scrutiny.

Manufacturing × Emissions and environmental monitoring × Machine learning · OS128 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
51 is the world moving	74 can we play, can we win	MEDIUM 0 named in the evidence	€309m observed evidence · per year	NOW budgeted procurement within 1...	L1 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying machine learning-based emissions monitoring to help energy-intensive manufacturers meet tightened EU ETS benchmark scrutiny with ultra-precise detection. It exists now because regulatory updates are increasing the need for precise data, and technology providers are launching ultra-precise monitoring systems, signaling a market shift towards higher accuracy.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€431m €77.6m – €2.3bn	€309m €55.7m – €1.7bn	€14.9m €2.7m – €79.8m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€1.0m €1.0m – €2.7m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Machine learning	3.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TML	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 3 declared geographies are outside the European reference data (TH, NG); the estimate covers the European footprint only.

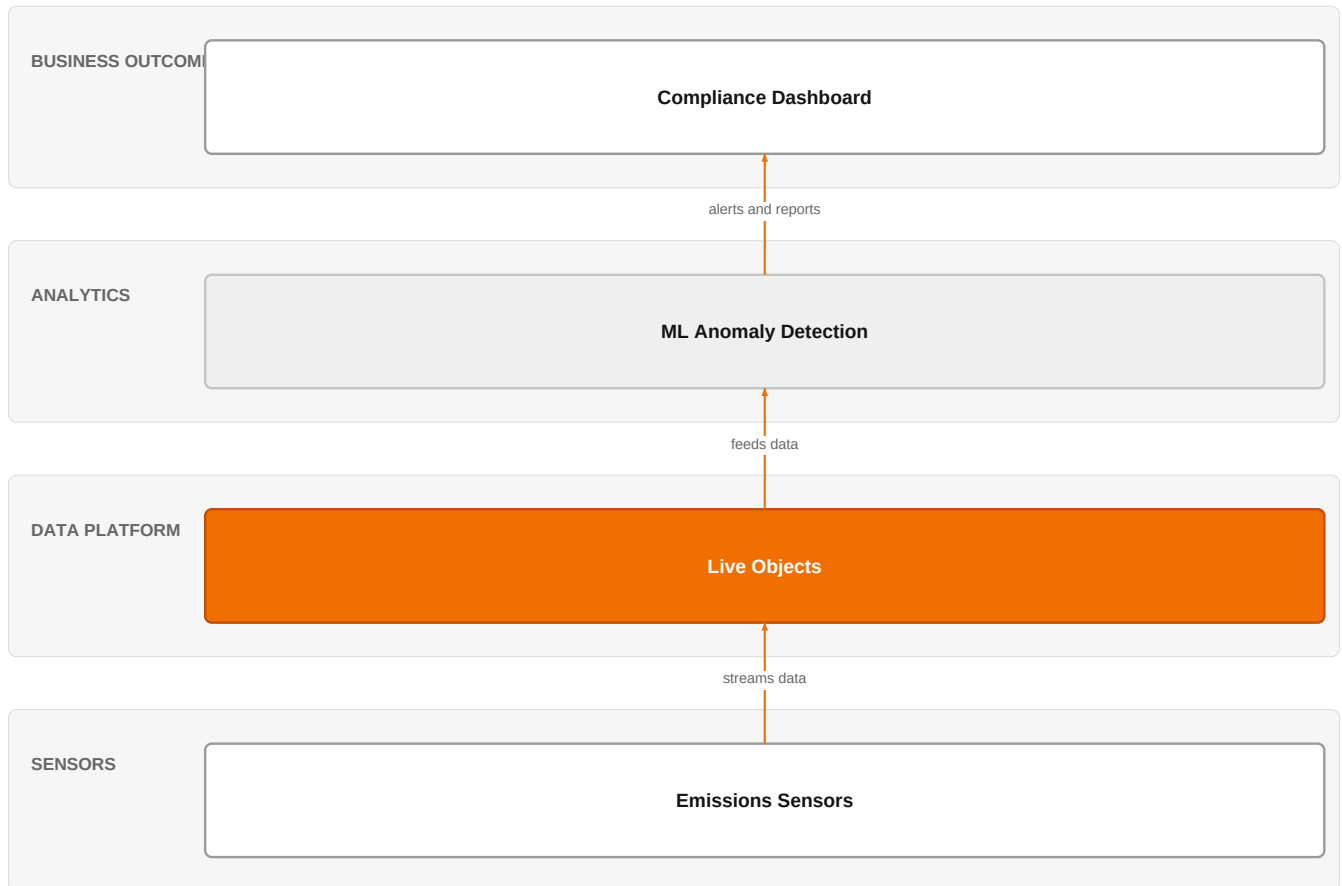
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	4.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 3 declared geographies are outside the European reference data (TH, NG); the estimate covers the European footprint only.

The solution, in one picture

Emissions Monitoring with ML



This diagram shows how Orange's Live Objects and ML capabilities turn raw sensor data into compliance-ready insights.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulatory updates are increasing the need for precise data, as seen with EU ETS benchmark scrutiny. Technology providers are launching ultra-precise emissions monitoring systems, indicating a market shift towards higher accuracy. Academic research is advancing machine learning for anomaly detection in industrial time-series data, which is directly applicable to emissions monitoring.

Sources: SIG-425F4DF1CB72 Envirotech Online 2026-06-20; SIG-D48860B13392 Manufacturing Today In 2026-05-20; SIG-03DDF2041F65 openalex.org 2026-02-03; SIG-8EE4F2310EA0 openalex.org 2026-02-01; SIG-EB5664191E1A openalex.org 2026-02-01; SIG-0E742BED215C openalex.org 2025-12-23

Who buys, and why now

The buyer is typically the head of sustainability or compliance at an energy-intensive manufacturer, while the plant operations manager feels the pain of manual emissions reporting and the risk of non-compliance. The trigger is the EU ETS benchmark update, which demands more precise emissions data, forcing them to upgrade their monitoring capabilities.

Sources: SIG-425F4DF1CB72 Envirotech Online 2026-06-20

Why it is hot now — every claim bound to a dated source

- Technology providers are launching ultra-precise emissions monitoring systems, indicating a market shift towards higher accuracy. [SIG-D48860B13392]

- Regulatory updates are increasing the need for precise data, as seen with EU ETS benchmark scrutiny. [SIG-425F4DF1CB72]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its AI/Data/Cloud experts, IoT experts, and Orange Group researchers, leveraging the Live Objects IoT platform for data ingestion. The engagement would start with a discovery phase to understand the customer's existing sensor infrastructure, then deploy machine learning models for anomaly detection on emissions data, and finally provide a dashboard for compliance reporting. The solution would be hosted on a cloud platform, possibly AWS or Google Cloud, and could incorporate NiCE for customer experience monitoring.

Why Orange can win

Orange can win by combining its certified infrastructure (ISO 27001, TISAX) with its expertise in AI, data, and IoT, and its experience with large industrial clients like Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass. However, the assets are thin in terms of a specific emissions monitoring offering, so Orange would need to differentiate through its integration capabilities and trust.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NiCE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include hyperscalers AWS and Google Cloud, who offer machine learning platforms and are also Orange partners. Atos/Eviden and Accenture are global systems integrators with strong manufacturing presence. NiCE, a partner, offers customer-experience platforms. Databricks, Dataiku, and Palantir provide data and AI platforms. Each competitor is strong in their respective areas, and the customer will compare based on platform maturity, ease of integration, and domain expertise.

Sources: SIG-D48860B13392 Manufacturing Today In 2026-05-20

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in OX: Smart Industries — also an Orange partner	structural	—

Google Cloud	Hyperscaler	sells Machine learning — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning	structural	—
NICE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning	structural	—
Dataiku	Data and AI platform	sells Machine learning	structural	—
Palantir	Data and AI platform	sells Machine learning; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How are you currently monitoring and reporting emissions for EU ETS compliance?
- 2 What specific benchmarks are you struggling to meet with your current data accuracy?
- 3 Do you have existing sensor infrastructure that could feed real-time emissions data?
- 4 What is your current process for handling anomalies in emissions data?
- 5 Who is responsible for compliance reporting, and what challenges do they face?
- 6 Have you evaluated any machine learning-based monitoring solutions yet?

Objections you will meet

They will say	You can answer
We already have a compliance system in place; why do we need this?	That's true, but the EU ETS benchmark update demands sharper scrutiny, and your current system may not provide the ultra-precise detection needed. Our solution can enhance your existing setup with machine learning to catch anomalies earlier.
This sounds expensive and complex to integrate.	We understand cost is a concern. However, our approach leverages your existing infrastructure and can be phased. We have experience integrating with industrial systems, and the long-term savings from avoiding penalties and improving efficiency can offset the investment.
How do we know the machine learning models will be accurate?	That's a valid concern. We use proven techniques from academic research and our own expertise. We'll pilot the solution on your data to demonstrate accuracy before full deployment.

Risks and unknowns

The main risk is that the customer may not yet feel the urgency to invest in advanced monitoring, or may prefer to rely on existing compliance methods. There is also uncertainty about the accuracy of machine learning models in detecting emissions anomalies under real-world conditions, and the integration effort with legacy systems. Additionally, the regulatory landscape may evolve, affecting the specific requirements.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Build a bid brief for an energy-intensive manufacturer's emissions monitoring tender, anchoring the technical solution on Live Objects and our AI/Data/Cloud experts, and citing Saint-Gobain Glass as a reference to demonstrate proven delivery.
Sales	In a meeting with a sustainability or operations leader at an energy-intensive manufacturer, say: 'We combine our Live Objects IoT platform with AI/ML expertise and ISO 27001-certified processes to help you meet EU ETS reporting requirements with the precision regulators now expect—and we've done it for clients like Saint-Gobain Glass.'
Strategist / Innovator	Commission a deep-dive brief on EU ETS benchmark tightening and map our AI/Data/Cloud and IoT experts against the data-acquisition and model-validation needs of energy-intensive manufacturers, using Saint-Gobain Glass as a reference for credibility.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D85076BA625E	2026-05-28	ted.europa.eu	1	Lithuania – Methane monitoring – (VPP-680) Gamtinių dujų (metano) nuotekiu matavimo, duomenų an...
SIG-03DDF2041F65	2026-02-03	openalex.org	1	Anomaly detection for industrial time series in process industry using informed machine learnin...
SIG-8EE4F2310EA0	2026-02-01	openalex.org	1	A Comparative Analysis of Machine Learning Models for Anomaly Detection in Industrial Smart Met...
SIG-EB5664191E1A	2026-02-01	openalex.org	1	A Systematic Review of Anomaly and Fault Detection Using Machine Learning for Industrial Machin...
SIG-0E742BED215C	2025-12-23	openalex.org	1	Improving Multivariate Time-Series Anomaly Detection in Industrial Sensor Networks Using Entrop...
SIG-1C6BA6D836AF	2025-12-17	openalex.org	1	FFT-Guided Multi-Window USAD with DTW–Isolation Forest for Reliable Anomaly Detection in Indust...
SIG-A320B94D827D	2025-12-06	openalex.org	1	Anomaly detection model based on unsupervised learning for multivariate industrial time series
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-6ADB457BBC36	2025-11-19	openalex.org	1	A Machine Learning Approach to Predicting Air Quality and Health Risks for Construction Worker ...
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-81F8AA75428C	2025-09-09	openalex.org	1	Time-Series-Based Anomaly Detection in Industrial Control Systems Using Generative Adversarial ...
SIG-4D1BCBC2C864	2026-07-14	Envirotech Online	2	Mercury emissions monitoring for environmental compliance - Envirotech Online
SIG-425F4DF1CB72	2026-06-20	Envirotech Online	2	EU ETS benchmark update puts industrial emissions data under sharper scrutiny - Envirotech Onli...
SIG-D48860B13392	2026-05-20	Manufacturing Today India	2	ABB drops ultra-precise emissions monitoring system - Manufacturing Today India

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:09+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.